```
#include <iostream>
#include <algorithm>
using namespace std;

int abs_i(int x)
{
    return x < 0 ? -x : x;
}

void fcfs(int req[], int n, int head)
{
    int total = 0, cur = head;

    cout << "\n--- FCFS ---\nSeek order: " << cur;

    for (int i = 0; i < n; i++)
    {
        total += abs_i(req[i] - cur);
        cur = req[i];
        cout << " -> " << cur;
    }

    cout << "\nTotal seek time = " << total << "\n";
}

void sstf(int req[], int n, int head)
{
    int r[100], done[100] = {0};
    int cur = head, total = 0;

    for (int i = 0; i < n; i++)
        r[i] = req[i];

    cout << "\n--- SSTF ---\nSeek order: " << cur;

    for (int c = 0; c < n; c++)
    {
        int idx = -1, best = 1000000;

        for (int i = 0; i < n; i++)
        {
            if (!done[i] && abs_i(r[i] - cur) < best)
            {
```

```
            best = abs_i(r[i] - cur);
            idx = i;
        }
      }

      done[idx] = 1;
      total += best;
      cur = r[idx];
      cout << " -> " << cur;
   }

   cout << "\nTotal seek time = " << total << "\n";
}

void scan(int req[], int n, int head, int disk)
{
   int r[102], count = 0, cur = head, total = 0;

   for (int i = 0; i < n; i++)
      r[count++] = req[i];

   r[count++] = 0;
   r[count++] = disk - 1;

   sort(r, r + count);

   int pos = 0;
   while (pos < count && r[pos] < head)
      pos++;

   cout << "\n--- SCAN ---\nSeek order: " << cur;

   for (int i = pos; i < count; i++)
   {
      if (r[i] == cur)
         continue;
      total += abs_i(r[i] - cur);
      cur = r[i];
      cout << " -> " << cur;
   }

   for (int i = pos - 1; i >= 0; i--)
   {
      if (r[i] == cur)
```

```
            continue;
        total += abs_i(r[i] - cur);
        cur = r[i];
        cout << " -> " << cur;
    }

    cout << "\nTotal seek time = " << total << "\n";
}

void cscan(int req[], int n, int head, int disk)
{
    int r[102], count = 0, cur = head, total = 0;

    for (int i = 0; i < n; i++)
        r[count++] = req[i];

    r[count++] = 0;
    r[count++] = disk - 1;

    sort(r, r + count);

    int pos = 0;
    while (pos < count && r[pos] < head)
        pos++;

    cout << "\n--- C-SCAN ---\nSeek order: " << cur;

    for (int i = pos; i < count; i++)
    {
        if (r[i] == cur)
            continue;
        total += abs_i(r[i] - cur);
        cur = r[i];
        cout << " -> " << cur;
    }

    if (cur != disk - 1)
    {
        total += abs_i((disk - 1) - cur);
        cur = disk - 1;
        cout << " -> " << cur;
    }

    total += disk - 1;
```

```
    cur = 0;
    cout << " -> " << cur;

    for (int i = 0; i < pos; i++)
    {
        if (r[i] == 0)
            continue;
        total += abs_i(r[i] - cur);
        cur = r[i];
        cout << " -> " << cur;
    }

    cout << "\nTotal seek time = " << total << "\n";
}

int main()
{
    int n, head, disk, req[100];

    cout << "Enter disk size: ";
    cin >> disk;

    cout << "Enter number of requests: ";
    cin >> n;

    cout << "Enter request queue: ";
    for (int i = 0; i < n; i++)
        cin >> req[i];

    cout << "Enter initial head position: ";
    cin >> head;

    if (disk <= 0 || n <= 0 || n > 100 || head < 0 || head >= disk)
    {
        cout << "Invalid input\n";
        return 0;
    }

    for (int i = 0; i < n; i++)
    {
        if (req[i] < 0 || req[i] >= disk)
        {
            cout << "Invalid request position\n";
            return 0;
```

```
    }
  }

  fcfs(req, n, head);
  sstf(req, n, head);
  scan(req, n, head, disk);
  cscan(req, n, head, disk);

  return 0;
}
```

```
siyasinnarkar@Siyas-MacBook-Pro-2 os % g++ disk_scheduling.cpp -o disk_scheduling
siyasinnarkar@Siyas-MacBook-Pro-2 os % ./disk_scheduling
Enter disk size: 200
Enter number of requests: 8
Enter request queue: 98 183 37 122 14 124 65 67
Enter initial head position: 53

--- FCFS ---
Seek order: 53 -> 98 -> 183 -> 37 -> 122 -> 14 -> 124 -> 65 -> 67
Total seek time = 640

--- SSTF ---
Seek order: 53 -> 65 -> 67 -> 37 -> 14 -> 98 -> 122 -> 124 -> 183
Total seek time = 236

--- SCAN ---
Seek order: 53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 199 -> 37 -> 14 -> 0
Total seek time = 345

--- C-SCAN ---
Seek order: 53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 199 -> 0 -> 14 -> 37
Total seek time = 382
siyasinnarkar@Siyas-MacBook-Pro-2 os %
```